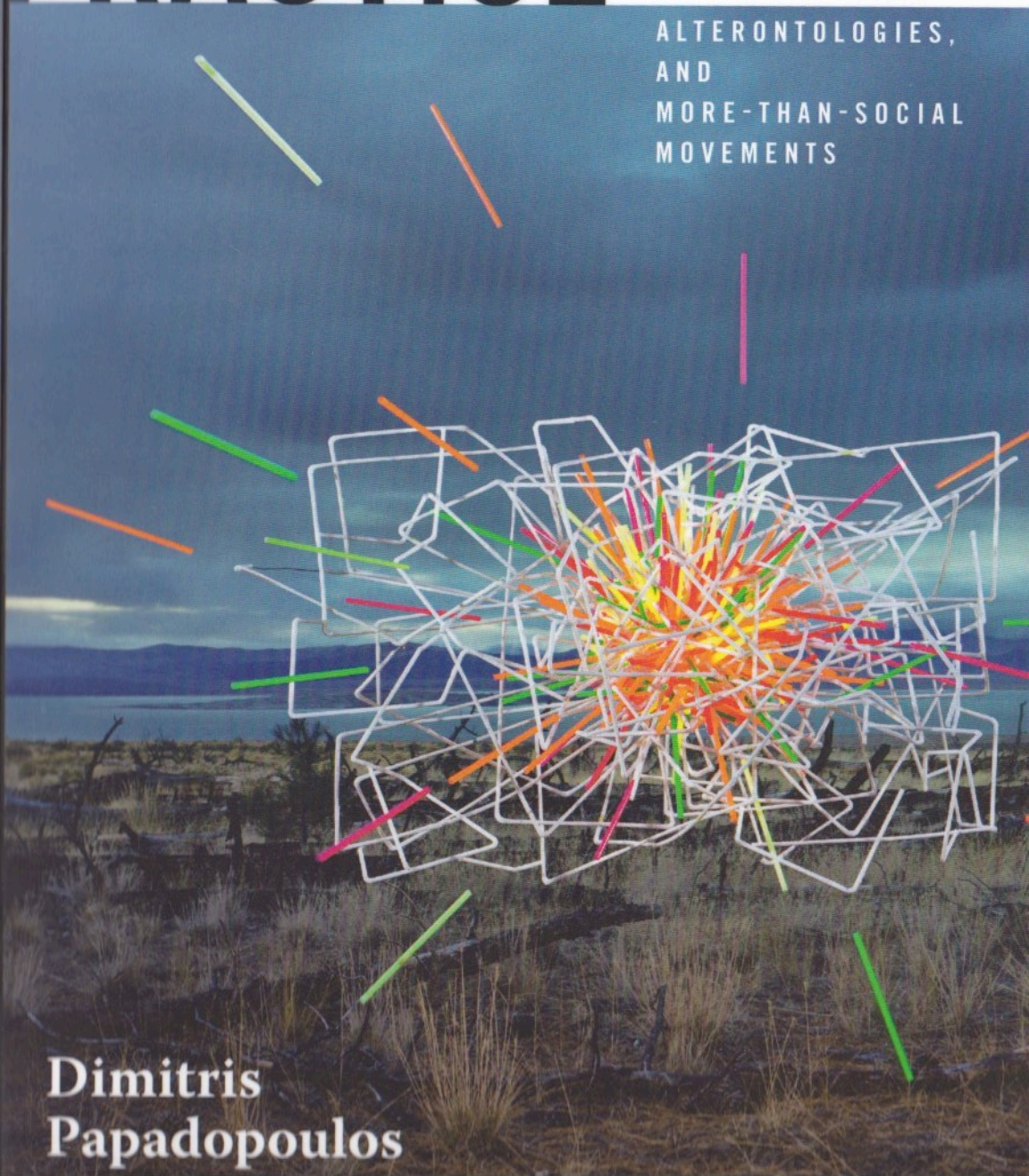


EXPERIMENTAL PRACTICE

TECHNOSCIENCE,
ALTERONTOLOGIES,
AND
MORE-THAN-SOCIAL
MOVEMENTS



Dimitris
Papadopoulos



**EXPERIMENTAL
FUTURES**

TECHNOLOGICAL LIVES,
SCIENTIFIC ARTS,
ANTHROPOLOGICAL VOICES

A SERIES EDITED BY

Michael M. J. Fischer and Joseph Dumit

EXPERIMENTAL PRACTICE

TECHNOSCIENCE,
ALTERONTOLOGIES,
AND
MORE-THAN-SOCIAL
MOVEMENTS

Dimitris Papadopoulos

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INTRODUCTION

THE TWO BEGINNINGS

This book has two beginnings. The first one is the decentering of the human in its relations to other species, machines, and the material world. What matters for this project is that the early twenty-first century saw the vision of "a more-than-human world," of "humans no longer being in control," and of a human-nonhuman continuum becoming slowly part of how we (some humans, that is) imagine ourselves (the human species) in the future. Let's hope that our species-being will be forever marked by this realization—although it may be too late.¹ I use the term *posthuman culture* to refer to this decentering of the human (and the humanist subject and its politics) into its relations to other living beings and the material world as well as the wider cultural realization of this decentering.² Technoscience has been the main force shaping posthuman culture: the continuous folding of science, technology, and the everyday into each other. Rather than the focus of this book, the histories and current formations of technoscience and posthuman culture constitute its first beginning, the stage on which its arguments are played out.

The second beginning is an affect rather than a phenomenon. A commitment rather than a thought. An obligation rather than an interest. A feeling of urgency to grasp the incapacity of the extraordinary social mobilizations that took place in countries across the North Atlantic and beyond since 2006 to instill social change.³ Seasoned social movements analysts

tell us that social movements have a *longue durée*—their effects are not clearly visible immediately, and what they achieve is often transposed in time and in different, often remote, fields of life. However true this may be, it is difficult not to feel that the mobilizations, struggles, and uprisings of the last ten years changed so many things and yet the transformational potential of these movements toward a materialization of socioeconomic and ecological justice was not accomplished. A map of this cycle of struggles would have many action points, sites of conflict, squares and plazas, linkages, transnational exchanges, alliances, and virtual meeting spaces,⁴ but I barely can count any broad effects in the direction of what these movements hoped⁵—knowing that it may be too early to look for these or to even have the conceptual tools and perceptual skills to see and grasp them. However, neither these mobilizations nor the eclipse of their claims for justice are the target of my analysis in this book. These social movements, their efforts, and their achievements shape the intellectual background and affective tonality of this book. I am gambling on a feeling with this second beginning.

This feeling tells me that there is a connection between the limited range of transformations that these movements have achieved and the displacement of the human and of human politics in posthuman culture. *Experimental Practice* attempts to investigate this connection. Consider, for example, the 2011 riots in Britain. They came unexpectedly. Speaking shortly after they happened, Paul Gilroy (2011) concluded by highlighting that many black communities drawn in the vortex of privatization and the intense neoliberal disintegration of British society are fragmented and often unable to defend and organize themselves.⁶ Later Gilroy (2013) expanded his analysis of the 2011 riots and attempted a comparison with the 1981 riots, emphasizing the very limited effects that the 2011 events had. The cry for change vanished soon after the riots stopped. They did not transform British society in the way this happened in the 1980s.⁷

I take Gilroy's diagnosis seriously and translate it to my words: these recent struggles show that there was no infrastructure that could hold together and protect the communities and perpetuate and multiply the effects of their actions. How can an ontology of community and infrastructures of communal connectivity be created?⁸ I am aware that the term *ontology* may be unexpected here. A notion of ontology will unfold across this book, but for now I mean the shared, durable, open material spaces—tangible and virtual—that can be inhabited autonomously by these communities.

AUTONOMY AFTER THE SOCIAL

Ontology and infrastructure are about something much greater than social relations. There are many social relations in our lives, probably more than enough, but there are not many material spaces where social and political autonomy can be performed. My bet in this book is that creating spaces of political autonomy and self-organization is not just a social affair. It is a practical and ontological affair that goes as far as to change the materiality of the lived spaces and the bodies, human and nonhuman, of communities. In fact, we may need to disconnect⁹ from the ubiquitous networks of social relations in order to create these autonomous material spaces of existence. If this is what needs to be done, then what is social movement politics today? What if we approach social movement action not as targeting existing political power but as experimenting with worlds? What if we see social movement action not as addressing existing institutions for redistributing justice but as the creation of alternative forms of existence that reclaim material justice from below? And, what if this becomes possible not when social movements engage in resistance to power but when they experiment with the materiality of life?

Experimental practice in this sense is about modes of intuition, knowledge, and politics that trigger intensive material changes and mobilize energies in ways that generate alternative and autonomous spaces of existence.¹⁰ Autonomy is meant as autonomous politics here, not as the modernist humanist value of individual independence and the seclusion of the personal and the private. The opposite is the case: autonomous politics requires material interconnectedness, practical organizing, everyday coexistence and the fostering of ontological alliances. And these are always more than human, more than social. They entail interactions, ways of knowing, forms of practice that involve the material world, plants and the soil, chemical compounds and energies, other groups of humans and their surroundings, and other species and machines.

The notion of experimental practice emerged as I was exploring if/where the two beginnings mentioned earlier meet. Bringing technoscience into the picture by retrieving the posthuman experimentations that are an undisclosed part of social movement action shows their politics under a new light: as *more-than-social* movements. I mean, as movements that do much more than just targeting visible and recognized social institutions; as movements that immerse into the human-nonhuman continuum and change society practically by engaging with both the human and the

nonhuman world. And they can do that only to the extent that they involve some part of technoscience. This is the ideal formula that this book seeks to advance: the metamorphosis of social movements to *movements*—that is, movements of matter and the social simultaneously. Movements of ontology. The book investigates this alternative perspective by slowly weaving the posthuman and technoscience into social movement politics and, the other way around, by weaving social movement politics into the practices of technoscience.

These arguments prolong work developed with Niamh Stephenson and Vassilis Tsianos in two previous books that attempted to show that creative social transformation since the 1960s and 1970s has been rarely the outcome of pure resistance or of opposition to power but of the remaking of everyday existence below the radar of control in mundane and yet unexpected ways.¹¹ Following the autonomous social movements at that time, we described how social conflict and social mobilizations drive social transformation instead of just being a mere response to (economic and social) power. In *Analysing Everyday Experience: Social Research and Political Change* (2006) we tried to show how this is possible at the level of individual subjectivity and experience. In *Escape Routes: Control and Subversion in the 21st Century* (2008), we tried to reconstruct this type of politics at a collective and community level. *Experimental Practice* closes the trilogy and addresses the same question on the level of matter. Experience, community, matter. The book seeks to put forward a form of politics that addresses these three aspects of our lives simultaneously: experimental practice.

BAROQUE FIELDWORKING

Making the case for an experimental practice of more-than-social movements is both empirically grounded in today's realities of social movement politics and simultaneously a deeply speculative undertaking. It is empirically grounded because it follows developments in the actions and politics of social movements that are already happening now. However, my aim is not to just follow and describe current social movements but to magnify specific aspects of their actions that can foster an experimental view of politics and point toward its transformative potential. Although all chapters attempt to contribute to this objective, they are all located in different fields and debates, have their own internal logic and argumentation, and

engage diverse methodologies and data. Introducing such a varying set of materials and methodologies allowed me to find a path for navigating the terrain that the mingling of the two beginnings mentioned earlier has created. As I look back to the years of work since I started doing research for this book in 2010, I feel that the methodologies I worked with and the materials I gathered and analyzed were almost imposed by the nature of the terrains I was delving into and the questions I was confronted with as I was moving through them.

Two of the chapters (3 and 8) present fieldwork studies: one is based on empirical materials from my long-standing engagement with migration and precarious work politics up to the end of the 2010s as well as fieldwork materials that were collected between 2009 and 2012 with my collaborator on this project, Vassilis Tsianos. The other one is based in my involvement in the maker and hacker communities across the East Midlands in the UK, primarily in Leicester and Derby and with different hacker and maker groups or activist groups that engaged with technological and ecological issues across the world. Chapters 6 and 7 present case studies that expand on the ontological and experimental implications of two cases of embodied technoscience that I have been investigating for several years now; the first one is on neuroplasticity, epigenetics, and the embodied brain, and the second discusses AIDS activism in the 1980s. Finally, the remaining four chapters collect and mix different materials: films, magazine and book covers, advertisements, historical accounts of specific events from secondary sources and different types of images, theories, Internet sites, scholarly texts, concepts, and science fiction literature.¹²

This approach allowed me to put together a speculative vocabulary of different experimental practices. Compositional politics, decolonial politics of matter, and alterontologies are some of the concepts I use to describe aspects of experimental practice from different angles. I would call the underlying methodological approach baroque fieldworking, a mix of politically engaged research, speculative historiography, and social science fiction. Although much of the work presented here is the outcome of different combinations of research and activism, this book is not an ethnographic account of this political participation. Rather, it is a theoretically motivated project grounded in sustained and lengthy political involvements as a committed practitioner and activist. Through these activities, the problems that I discuss in the book were presented to me and then took shape and form. One could even say that the whole book is an attempt to negotiate problematics that arose through these practical and political involvements.¹³

One of the aims of this book is to explore different narratives of emergence of experimental practice. This necessitates engaging speculatively with the historicity of the topics and events that are central for my argument: I bring together different, often discrepant, historical events, artifacts or chronotopes in order to re-create a possible historical trajectory that unearths a present phenomenon that has not fully emerged yet. I try to explore the otherwise, the not yet fully materialized, the unknown from significant proximity. I am less interested in exploring the unknown from distance or even in exploring the known by creating (critical) distance through a predetermined methodology. I am drawn to getting as close as possible to something that is already there and yet has not fully emerged. And this involves getting as close as possible to its past by unlocking its speculative potential in order to reconstruct promising alternative histories and virtual futures.

The combination of empirics and theory and the development of a speculative historiography are part of my attempt to emphasize the fictional side of social science writing. I discuss this in length in chapter 2 but for now it may be important to say that rather than trying to bring fictional tropes, contents, and genres into social science, the social science fiction I am pursuing here attempts to write social science *itself* as fiction. That is, the objective is to write social science in a scholarly social science and social theory fashion (by following standard citational and stylistic conventions and by docking onto existing debates) and to incorporate in it, almost imperceptibly, the fictional and speculative dimensions that emerged through my political and practical engagements.

Social science fiction helped me to elevate something that is happening already but is still not a defining moment of social movements—that is, experimental practice—to a form of politics that is forcefully present in our realities. I see social science fiction as 100 percent empirically grounded and 100 percent fictional. It is both at the same time. It is almost as if there is a spiritual dimension in experimental practice, a “material spirituality” (Puig de la Bellacasa, 2014a), that this process tries to reveal in our realities. The implicit philosophy of this project is therefore neither realist nor social constructivist, neither critical realist nor a constructivism without adjectives either. It is an approach as experimental as the worldings it engages with require it to be.¹⁴ Thus, despite the intensive engagement within the fields I was involved in, I did not have a predetermined methodology and I did not *do* fieldwork. Rather, I was, as are many other humans, politically active *in* these fields for reasons that are far greater than this study. I was (and still am in some cases) fieldworking.

Instead of trying to create some unity and permanence while putting together the materials, concepts, and ideas that I gathered while fieldworking, I have developed a baroque piece containing a transitory, often expressively dissonant and extensively ornamental main text folding into an infinitely cavernous, continuously curvilinear and often disjunctive set of relations to other texts and thoughts presented in the notes of the book.¹⁵ In this sense, I constructed a transfigured objective baroque world where speculative occasions, spiritual meanings, empirical materials, and social research have equal place.¹⁶ The aim of baroque fieldworking and social science fiction is to present a world of abundance—and I see this not only as a stylistic attempt but as a political obligation of the text.¹⁷ Instead of dearth, which is the outcome of traditional management and control, I looked for abundance as the outcome of material self-organization and autonomy and, of course, of experimental practice.

ESCAPING HUMANITY (OVERVIEW)

I trace experimental practice in different fields of life and through different cases and occasions. In all of them I try to retain those aspects that allow me to understand the material practices of social movements in their quest for justice. These practices that fuse justice and ontology will allow me to unearth the more-than-human and more-than-social aspects of social movements and their slow path to become *movements*. I start this journey in chapter 1, “Decolonial Politics of Matter,” which serves as the theoretical introduction to the book and provides a conceptual diagram of its key questions and arguments. The chapter argues that in order to be able to address questions of justice, social movements engage in a decolonial politics of matter: they learn and experiment with changing—literally—the material composition of life in ways that delink from the Western epistemic appropriation of matter as an open frontier for exploration and enclosure.

Chapter 2, “Biofinancialization as Terraformation,” and chapter 3, “Ontological Organizing,” make up the first part of the book and attempt to grasp the role of these experimental material practices in contemporary social movement politics. Chapter 2 aims to establish why the experimental moment in social movement politics is necessary and, indeed, unavoidable. The chapter describes the ascent of financialization since the 1980s, which brought with it a culture of valuation that spread well beyond financial markets and came to pervade everyday activities, subjectivity,

ecology, and materiality—biofinancialization. I argue that the assetization of life and the financialization of bios has made the current regime of production and accumulation untouchable in political terms. This, of course, poses significant challenges to traditional politics, including social movement politics, because it interrupts established channels for social change through existing political institutions. The question then is how to imagine the autonomy of politics when biofinance becomes molecularized in code and in matter.

Chapter 3, "Ontological Organizing," approaches social movement action from the reverse perspective: from the practices of movements on the ground. In this chapter I work with materials from a long fieldwork study of migration activism and migration movements developed and analyzed with my collaborator Vassilis Tsianos. Here we argue that many social movements today, and certainly the migration movement, increasingly change the ways they perform politics. They avoid targeting directly institutions of power and organize outside existing political channels by setting up alternative ways of being that support their aims. In the case of migration, this means the creation of imperceptible but durable infrastructures and ontologies of existence that facilitate the freedom of movement of migrants. The sovereign regime of mobility control is displaced on the level on which it attempts to take hold: the everyday movements of transmigrants. The securitization of borders and spaces is challenged by organizing common ontologies of existence below the radar of pervasive political control and by creating alternative everyday worlds: the mundane ontologies of transmigration, the *mobile commons* of migration.

Chapter 4, "Activist Materialism," and chapter 5, "Insurgent Posthumanism"—the second part of the book—mix and remix different historical incidences, conceptual resources, and perceptual strategies in ways that allow me to trace the experimental dimension in historical forms of political mobilization and social movement action long before anything like posthuman culture and technoscience existed. Chapter 4 specifically discusses the adventures of the concept of materialism and its uneasy relation to political activism. Here I explore possibilities for enacting activist interventions in conditions where materialist politics is not primarily performed as a politics of institutions but as the fundamental capacity to remake and transform processes of matter and life. What is activism when materialism is not about a politics of history but a politics of matter? What is materialism when it comes to an activist engagement with matter itself?

Chapter 5, "Insurgent Posthumanism," traces experimental practice as the posthuman making and remaking of alternative material conditions of existence in different historical cases of social movements and theoretical accounts of politics. But coupling politics and experimental practice and fusing social movements and the posthuman defies much of posthumanism's current assumptions as well as many of the theoretical presuppositions of the politics of social movements. The chapter gradually weaves together politics and posthumanism along three distinct experimental practices: the making of common material worlds (not just a common humanity); the embodiment—literally—of radical politics; and, finally, the enactment of justice through a materialist, nonanthropocentric history.

Chapters 6, 7, and 8 constitute the third part of the book. They bring back the discussion of experimental practice to the present by performing the rapprochement between social movements and the more-than-human worlds of technoscience. How do we perform experimental practice today? If the experimental cannot be thought independently of technoscience, the aim of this third part of the book is then to explore different ways to conceive politics in technoscience. As I argue throughout this part, the experimental practice of more-than-social movements is enacted with, within, and occasionally against, but never outside, technoscience.

Chapter 6, "Brain Matter," is a case study of the notion of neuroplasticity and the emergent epigenetic nature of the brain. At the heart of the vision of plasticity circulating equally in popular culture and in the sciences of the brain lies the possibility of recombining brain matter and understanding the making of ecologically dependent morphologies in a nondeterminist manner. But plasticity as recombination is not only a radical challenge to predominant determinist assumptions of the brain, it becomes also one of the major avenues through which politics becomes articulated within neuroscience. The chapter explores different prevalent versions of such politics related to neoliberal markets, processes of governance, and traditional visions of social liberation. Engaging with these different forms of politics allows me to explore their specificities and, indeed, limitations in establishing questions of justice in the relation of humans to their brain. Here I ask what an experimental practice of the brain is or would look like and introduce the notion of composition as a key feature of experimentation with matter.

Chapter 7, "Compositional Technoscience," picks up the idea of composition and proposes to reexamine how politics has been conceived within science and technology studies: the politics of credibility and expertise, institutional participation, the governance of human-nonhuman relations,

and the inclusion of marginalized experiences. I debate these different approaches to politics through a case study of AIDS activism in the 1980s and show that more than an organized response of the gay community targeting inclusion in existing institutions and expert committees, AIDS activism was the product of the community's efforts to survive the epidemic and the prevalent social hostility. In fact, the community did not preexist these efforts; it became a community only by engaging in an extremely wide spectrum of everyday material experimental practices that made its very existence possible. I refer to these experimental practices as compositional: the creation of alternative forms of life that primarily allow a certain actor to be able to exist, to articulate with other actors and forms of life, and to address questions of justice by changing its everyday material conditions.

The last chapter of the book, chapter 8, "Crafting Ontologies," presents a larger fieldwork study of the maker and hacker culture in the British East Midlands and beyond. The main thesis of the chapter is that the achievement of invention in technoscience takes the form of dispersed experimentation within more-than-human and more-than-scientific worlds: distributed invention power. In maker and hacker spaces, the focus of this chapter, we can trace how distributed invention power is organized: the topological stacking of materials and processes; ecological transversality and the emergence of new compositions of code and matter; the proliferation of commensal relations between the participating actors; the complex traffic between instituted and community technoscience; the involution of human, animal, and inorganic actors; the centrality of craft and experimental labor in makers' ethopoiesis; the precarization of work and multiplication of free labor; and, finally, the continuous folding of the commons and the private and public spheres into each other. What is constitutive of these diverse practices of making and other movements of community technoscience is that they change the conditions of knowledge production by recomposing the fabric of everyday life: the stacking and forking of worlds into alternative ontologies. What is at stake here is not just technoscience itself but the ontological constitution of life and the attempt to defend it. In posthuman conditions, traditional politics and the corresponding social movements can support us in this endeavor only to a limited extent. The alterontologies of more-than-social movements do not just represent a new form of political organizing. Something else, something existential is at stake here: alterontological politics is a possible way to survive a world that is disintegrating through human action. Alterontologies may be a way to escape humanity.

DECOLONIAL POLITICS OF MATTER

ONTOLOGICAL POLITICS

The rise of technoscience in its contemporary configuration since the 1960s and 1970s resonates with a cultural imaginary marked by the idea that social transformation is primarily driven by material transformation.¹ Technoscience creates new ontologies; it is world-making and history-making.² The simultaneous production of society and ontology is often described as ontological politics: there exist multiple ontologies rather than just one, and these different ontologies are enacted by the actors involved in them.³ As I discuss later in the book (especially in chapters 7 and 8), this is not an epistemological question, it is a practical one. Depending on the specific actors involved, ontologies are practiced differently and thus are materially different as such. This is a "multinatural" world.⁴ And it is ultimately a political question which ontologies a certain actor participates in, and how. Politics here means that by performing ontology in a single concrete way rather than any other, we change the very constitution of being and its material organization in a specific direction. Ontology is not a description of the state of things, but of ways of being that include alternative possibilities of world-making.

What absences and what silences are produced when we conceive our engagement with and within technoscience as ontological politics? And what ways of being and acting, what voices does this understanding of technoscientific politics privilege? Along with Boaventura de Sousa Santos (2001) I am interested in the study of absences. Rather than exploring what ontological politics is or how it is theorized, I aim to discuss what ontological politics actively produces as absent and nonexistent.⁵ Is the politics of (more-than-social) social movements that I mentioned in the introduction a form of ontological politics? If not, what are the alternatives? As de Sousa Santos (2004, p. 178) writes, "There is no single, univocal way of not existing. The logic and process through which hegemonic criteria of rationality and efficiency produce the non-existence of what does not fit them are various. Non-existence is produced whenever a certain entity is disqualified or rendered invisible, unintelligible, or irreversibly discardable. What unites the different logics of production of non-existence is that they are all manifestations of the same rational monoculture." With Susan Leigh Star (1991) we can think of the production of the nonexistent as an accumulation of residuals created in the process of making the world: residuals produced through work that has been invisibilized. What are the absences, the residues, the invisibilized labors that cannot be considered in ontological politics? What is rendered silent in the process of performing ontological politics?

THE FRONTIER OF MATTER

In ontological politics, matter is not just raw material for other social or political ends; rather, matter itself is opened up as a space of expansion.⁶ Matter in all its expressions and formations is not a substance with fixed qualities and given potentials; matter is active and creative, complex and enlivened. The shift to the ontological denotes an interest in immersing into this process of self-ordering and in co-acting within it—to let other possible ontologies emerge.

Consider, for example, the nascent field of epigenetics. Over the past decade, a new set of theories and experiments exploring the unpredictable dynamics of gene expression has begun to take center stage in genetics research. The *epigenome* describes the overall state of a cell in flux, each point in time yielding multiple cascading possibilities for divergence of individual phenotypes. The inert genome is thereby supplemented by

a softer, more adaptable, epigenome, incorporating mechanisms capable of responding to the environment, sensing "time" and retaining "memories" that regulate subsequent development. Within myriad microscopic epigenomes, the effects of the wider social and physical environment are translated via biochemical interactions to become an integral part of a fluctuating landscape of gene expression.

I discuss epigenetics in chapter 6 (for an extensive discussion, see the work with my collaborators on this project, Emma Chung, John Cromby, Chris Talbot, and Cristina Tufarelli)⁷ but for now I want to highlight that the epigenome as an emerging object of study as well as a theoretical framework (epigenetics) could be described as a set of alternate ontologies that are made by a multiplicity of actors who contribute to the field from many different and often conflicting perspectives and positions. Their specific ways to do research are not primarily shaped by their commitment to some external politics to the field (such as left or right, liberal or emancipatory, or conservative or progressive), not least because many of the actors involved are nonhuman others and do not have such politics. The politics that one can find are intrinsic to epigenetic research, yet their connections go far beyond the field of epigenetics and address broader questions of politics and justice.

I wonder what the ontologies of the epigenome will look like in the next decade when this nascent field will start to take a more definite shape. Which ontologies of the epigenome will develop further, and what will they look like? Which will disappear? How and why? Ontological politics captures this process. Performing ontological politics means to open up matter as a field of exploration, experimentation, and, ultimately, appropriation. Ontological politics conceives matter as a *frontier*. In every frontier, expansion takes place as inclusion of new territories and entities in a process of continuous creation of new worlds.

COLONIALITY AND PRODUCTIONISM

Every frontier has a promise: to liberate the one who moves into the open space from the limitations that preoccupy life before the frontier opens. The promise of the new frontier of matter is to liberate material action from being dominated solely by social imperatives—class, sexuality, race, power, religion, culture, inequality—and to develop a radical commitment to matter (and to its multiple expressions). But simultaneously every

frontier has a secret: in order to expand and include new spaces, entities, and actors, it needs to exercise power over them and silence or oppress some of them in order to make them fit. Ontological politics on the one hand expresses this liberating move—that is, the openness of ontology to co-action of all actors involved and to the multiple possibilities within it—and on the other hand administers its secrets: How many actors can be included? How? What is going to happen with the rest of them?

Every moving frontier is contemporaneous with a form of liberation, and simultaneously it enacts oppression. In the American frontier, the laborers who escape the wage labor market of British America to become independent peasants and artisans from the seventeenth century onward move into the frontier to the West only to bring savagery, destruction, and the dispossession of land.⁸ In the Eastern Cape frontier of South Africa, the opposition of the Afrikaners to British colonial power and their subsequent migration came only to consolidate militarized white supremacy and extreme nationalism.⁹

The frontier opens as moving people (most often violently) appropriate new territories and then include and integrate them into their realities. And as people move, the new territories absorb them into their workings and transform them. The logic of every frontier is the logic of inclusion, the colonial inclusion of hinterland and the outside into some form of control that spreads well beyond the actual frontier itself. The term *coloniality of power* describes an order of power that emerged as a result of colonialism and colonial administrations but survived their later demise to structure culture, knowledge, economy, and our ways of being.¹⁰ Colonialism shaped modernity to such extent that modernity itself would be impossible without it: modernity/coloniality. And with it also our epistemic practices and systems of knowledge are constituted by the modernity/coloniality nexus. The frontier of matter, this newly opened frontier that operates as an epistemic appropriation of matter, exists within the practical logic of the coloniality of power. The frontier of matter operates within “modernity’s epistemic territory,” inescapably.¹¹

In the movement of the frontier, territory is considered to be a space on which nobody has property rights and therefore it is open for appropriation, *terra nullius*.¹² It is only because of this rendering of the territory as unclaimed that the frontier legitimizes its moves and the erasure of already existing systems of knowledge, local epistemic traditions, and the communities that sustain them. In modernity’s epistemic territory, mapping

space, representing its entities, and shaping new political institutions in order to include these entities (and exclude and, often, erase others) is the way the frontier expands. Through these processes the frontier is turned *productive*, in a double sense: the space of the frontier is turned into a space of production, that is, a space that is produced through relations of forces and power as well as a space that is gradually docked to existing systems of the production of goods, knowledge, and commodities.¹³

The appropriation of the frontier is a double act of organization: it is organized as a political space through processes of representation, and it is organized as a productive space by turning it to an accumulation system. Inclusion is about enclosing the new spaces into a regime of representation and a regime of accumulation that sustains the power of coloniality.¹⁴ So also in our case: when matter becomes a frontier, the attempt is to make it productive—politically productive—as well as render it compatible with the existing mode of production. Ontological politics are these specific practices that perform the inclusion of new formations of matter into the accumulation regime of current economies; I discuss specific ways that this happens in chapter 2 and again in chapter 7. As the frontier of matter moves, its political institutions also move. This necessitates the creation of new political forums to accommodate these emerging formations. Including nonhuman actors into polity is about rendering them amenable to being represented;¹⁵ as they become representable, that is, as they become identifiable within the coordinates of an existing mode of political organization, they pressure existing political institutions to change and to accommodate these new actors.

Ontological politics is a description of practices that portray how the frontier of matter advances and that unfold within spaces marked by colonialism. I am not saying that ontological politics is a colonial enterprise (although this may be true in many cases). What I am saying is that ontological politics exists within the modern epistemic territory that is constituted by its coloniality: ontological politics makes the frontier of matter, modernity’s ultimate frontier, move. Thinking politics in technoscience with ontological politics allows us, as I discuss in chapter 5, to challenge widespread anthropocentric understandings of what matter is and to introduce the idea of the coexistence of multiple contingent possibilities that emerge in the process of movements of matter. Ontological politics describes the forces exercising pressure on the outer limits of constituted political institutions to differentially include more human and nonhuman actors, relations, and ontologies in a given political configuration in order

to allow for these ontologies to emerge and become embedded in our social and material worlds.

The epigenome, a neuron, an MRI scanner, or a new subsea fiber-optic cable are not just actors that are differentially enacted by existing social institutions and people to create new forms of existence and new ecological configurations; they are also embedded in actual epistemic and value production processes that maintain the existence of the frontier.¹⁶ In other words, ontological politics is politics performed in order to keep the frontier of matter moving by activating new material processes, allowing multiple possibilities, including new material actors, representing them, assigning them specific rights and positions, inserting them into the political sphere, extracting value from them, and speculating on their value to multiply financial yields. Ontological politics is not just a theoretical tool depicting politics within technoscience, but also the *modus operandi* of rendering matter productive, in the double sense of productionism: politically representable as well as creating value and innovation.¹⁷

MATTER AND JUSTICE

What would it mean to question the implicit coloniality of the frontier of matter? To what extent can material actors object to their enlisting in the productive machine of the frontier? Is it possible to think of actors escaping modernity's epistemic territory? How can the epigenome object to specific meanings and functions that it is expected to perform in the emerging research on epigenetics? Can a transatlantic fiber-optic cable challenge its specific use by a certain consortium of companies that maintain it and exploit its labors? What labors do these actors do that remain nonexistent, absent, invisible while many other of their labors are fully absorbed in the productive regime of the frontier? As I argue throughout this book, in order to be able to start looking for answers to these questions we will need to challenge the assumption that the *inclusion* of these actors in constituted political institutions will securely bring their absent voices and their invisibilized work to light.

With Star (1983, 1991) I am interested in the organization of the invisible and erased work that remains hidden when ontological politics is performed, rather than the organization of visible work as such (and this pertains to human and nonhuman actors alike, as I discuss later in the book). Erasure and inclusion coexist in the modern/colonial epistemic territory. Rather

than inclusion, then, in the chapters that follow I aim to explore alternative political practices that attempt to disconnect and redraw the conditions and terms of this epistemic territory, "a delinking that leads to de-colonial epistemic shift and brings to the foreground other epistemologies, other principles of knowledge and understanding and, consequently, other economy, other politics, other ethics" (Mignolo, 2007, p. 453).

Following Jacques Rancière (1998), my starting point is not inclusion but the emergence of the invisibilized and the imperceptible, of those who have no place within existing normalizing political institutions. I understand politics here as a deep dispute over the existence of those who have no part in constituted institutions or even of those who refuse to participate in them.¹⁸ And this form of politics within the frontier of matter happens when those who have no part change the material conditions of existence in a way that cannot be overheard or simply included in existing political institutions.¹⁹ In chapter 3, for example, I discuss this in the case of people's migration and in chapter 7 in relation to AIDS activism. I am thinking with Starhawk (2002) of a form of politics that is not exercised as the *power over* a territory or simply a power that appropriates what is within it; rather it is politics as *power with*, the power of creating alternative common forms of life that reorder the language and practices of existing political arrangements of constituted power.

Instead of asking questions such as "How can we include different humans and nonhumans in our institutions?" "What is matter in itself?" or "Who is not included?" I focus on how actors create alternative ecologies of existence that become inhabited by these silenced and absent others, by those who have been rendered residual and invisible.²⁰ Rather than exploring with ontological politics the making of different ontologies, I am interested in a *decolonial politics of matter*: politics that, by instituting direct changes on the material level of existence, challenge existing conditions of inclusion *and* the idea of inclusion that rests on epistemic coloniality. Deleuze in his homage to François Châtelet points out: "No science, but rather a politics of matter, since man is entrusted with matter itself" (Deleuze, 2005, p. 717). This is a politics of matter not because humans are in charge of matter but because certain groups of humans and nonhumans can continue to exist only to the extent that they develop other alternative entanglements with matter. Matter is hope. Ontology *is* desire (see chapter 4 for further discussion).²¹

Beyond ontological politics as a general description of politics within the frontier of matter, I want to think of a decolonial politics of matter

as a committed politics to make alternative worlds of existence with and against and outside the productionist moves of the frontier. But unlike traditional decolonial approaches that I discussed earlier in this chapter and throughout this book, my approach here is not primarily the epistemological unsettling of modern/colonial knowledge production. The unsettling of the frontier of matter can only take place on the material level, and it can only be a practical ontological question. I am less interested in how many ontologies exist and whether it is possible to create a pluriversal world (something that I discuss in length in chapter 8); I aim to explore practices that destabilize, delegitimize, and, ultimately, decolonize the frontier of matter through the alternative production of materiality: alternative ontologies. This decolonial politics of matter attempts to restore justice step by step through everyday practice (something that I discuss in chapter 5). In politics of matter, justice becomes ingrained in the materiality of being: in the soil, in the water, in our bodily tissues, limbs, organs, cells, genes, and molecules. Decolonizing settler land is one thing; decolonizing matter is another (even if they are tightly connected).

Material justice is a form of justice that happens even before its epistemological representation and political inclusion into constituted governance has taken place. This creation of alternative ontologies characterizes the actions of what I refer throughout this book as *more-than-social* movements: those that, rather than aiming at social and political power, primarily change the immediate ontological conditions of life. This book provides different materials—historical (chapter 4 and 5), empirical (chapter 3 and 8), and conceptual (chapter 2, 6, and 7)—that allow us to grasp how classic social movements can be conceived as more-than-social movements.

We are somewhat trained to believe that inclusion comes first: that inclusion in structures of social power is what politics is about. It is probably the other way around: politics is when certain actors, imperceptible actors, emerge in the political scene and change the very constitution of being by—literally—materializing ordinary relations of justice; inclusion can only follow this move. Instead of prescriptive justice, I am searching for material, processual, and generative justice,²² which, rather than being focused on normative issues, is concerned with fusing justice and matter: practical justice, thick justice.

ALTERNATIVE FORMS OF LIFE

How does thick justice materialize? How can justice be installed through bodies, with other animals, things, and matter? How can we think matter and justice in a nondualistic manner? Let's turn upside down Clifford Geertz's "thick description" (1993): thickness for Geertz is semantic, let's seek material thickness; practice for Geertz is text, let's read practice as the material worlding of existence. Thick justice is about reclaiming materialist politics in the age of technoscience by making alternative ontologies: alterontologies (a concept that I discuss throughout this book and in particular in chapter 7). When ontological politics goes to formal institutions, politics of matter goes to the everyday: the space where alterontologies are crafted. Alterontologies = alternative forms of life.

I borrow the term *forms of life* from Langdon Winner (1986, especially chap. 1), who traces it back to Ludwig Wittgenstein as well as to Karl Marx. In forms of life we encounter a reweaving of the social and the material through the development of new practices, knowledges, and technologies. A practice, a set of practices, a device, a new form of connection becomes part of a form of life *by changing it*. There are no users, no tools, no symmetrical representations, no assemblages, no networks. There are just forms of life that set up the material constraints to what we are, what we can become, and how we co-construct each other. Wittgenstein (1958, p. 226) writes: "What has to be accepted, the given, is—so one could say—*forms of life*." I understand this as the making and sustaining of forms of life that have to be accepted because they transform the material order in ways that cannot be bypassed or neglected. Every social context, every sociotechnical environment, every ecosystem has enough space for alterontologies: conflicting alternative forms of life.

In previous work we called the politics of these social movements *imperceptible*.²³ Imperceptible politics are not invisible, yet they neither aim for their inclusion in constituted institutions nor claim visibility in the existing regimes of polity; rather, they transform the immediate conditions of existence without having as their central target their own representation in main political institutions. Imperceptible politics is the creation of new speculative figurations, new deliberate actual constructions, which put us right in the heart of new material and experiential forms of life. Günther Anders's call to train our capacity for "moral fantasy" and Walter Benjamin's "speculative experience" are behind these ideas. Anders discusses the inadequacy (what he later develops as a philosophy of discrepancy)

between our feelings and the unforeseeable effects of things and demands that we train the elasticity and capacity of our imagination (Anders, 2002, p. 271ff).²⁴ Benjamin (1996b) discusses also the magical and spiritual language of things and develops the idea of speculative experience as a means of recognizing the wholeness of life beyond a naïve utopian idealism or blunt versions of materialist dialectics (I discuss this in chapter 4).

The quest, then, is how speculative experience becomes materialized in the making of alternative forms of life and simultaneously escapes the pressures of productionism—in its double sense as discussed earlier: inclusion into the political representation and into the current regime of accumulation. Instead of seeing technoscience as the instance against which claims need to be addressed and instead of seeing that claims of justice need to be imported from outside into technoscience, the aim is to, literally, *craft* alternative words. Crafting—that is, the everyday making of alterontologies—is one of the main themes that this book tries to explore. In chapters 4 and 5 I discuss different historical instances of making alternative ontologies of existence, and in chapters 6, 7, and, in particular, chapter 8 I collect different aspects that delineate crafting within contemporary (more-than-)social movements.

DECOLONIZING CRAFT

Craft is the result of a skill developed to such a degree that what matters is not the skill itself but the very moment of making. Craft is an excess of practice, “the desire to do a job well for its own sake,” as Richard Sennett (2008, p. 9) writes. Craft and artisanal production were always in the heart of making things; indeed, they defined the birth of modern science until its gradual absorption into big science.²⁵ Craft in this picture becomes a necessary prerequisite and the absent mediator of what Barad (2007) calls an intra-active relation with matter. However, this tight interconnection between craft and the modern epistemic territory reveals also its limits, the need to approach craft from a decolonial perspective. Craft and artisanal skills are the invisibilized labors, the erased residuals that sustain any situated relation to matter. Simultaneously, craft is inescapably located within the frontier of matter. This specificity of craft defines the approach to decolonization that I use throughout this book. It is not primarily about the decolonization of seized lands, ways of being, and Eurocentric epistemologies or the creation of a pluriversal world. All of these are surely important

parts of the decolonial project. But my attention is on something less central but crucial for my specific project here: the decolonization of our relation to matter and materiality from a position situated in northern Europe, where I live and I am politically active.

This relation to matter and materiality within northern Europe and more broadly the North Atlantic is one that is mediated in multiple ways through technoscience, the contemporary inheritor of scientific practice of the modern epistemic territory. Decolonizing matter means, then, decolonizing technoscience and its absorption into the frontier of matter. However, I approach this decolonizing project not by deconstructing technoscience from a perspective that would, for example, reveal how technoscience helps to perpetuate the coloniality of power—how it is used to sustain coloniality in geopolitical relations, how infrastructures maintain social power over indigenous communities, or how technoscientific knowledge reproduces colonial relations by other means, to name just few examples. Instead of approaching technoscience from the outside as an object of study—and eventually as an object of critique—in this book I rethink the practices of technoscience *itself*, hoping that this investigation will contribute to such broad decolonial projects. To what extent is it possible for social (or more-than-social) movements to abandon the split between politics, justice, and technoscience that pervades the ontological politics of the frontier of matter? I interrogate craft as the moment where the bifurcation between technoscience and politics, matter and justice can be potentially abolished. Inasmuch as craft is an inheritor of the colonial architectures of the modern epistemic territory, it is also a practice that can contribute to its destabilization.

Consider, for example, how grassroots ecological activism has contested the externalization of the costs of production to the environment by crafting a multiplicity of alternative forms of life (even if the relations between these different forms of life may conflict at times): repatriations of indigenous land, revegetation, biodynamic principles of farming, water conservation, inner-city food gardens, recuperation of traditional and indigenous systems of land use and land care, cooperative production, organizing against extractivism and the agroindustrial frontier, creation of alternative seed banks, permaculture activism, soil regeneration, whole-farm organization, urban gardening, ecofeminist advocacy, bioremediation projects, disruption of agribusiness, open-source agriculture, experimentation with biofuels, reclaiming of dispossessed spaces, production of alternative research, making of alternative collectives, setting up local

systems of exchange and transactions, the early years of Seeds of Change, Earth Activism—all examples of crafting alternative material-ecological justice on the ground, multiplying livable worlds, or making alterontologies. The political organization of social movements here does not pre-exist the making of alternative forms of life; rather, political organizing *is* the crafting of alterontologies. Ontological organizing matters within social movements (I discuss this in detail in chapter 3). Ecological activism of the kind I describe here has addressed ecological destruction and the ontological politics of technoscience not primarily by opposing it or by exposing its connections to big agribusiness but also by occupying its very activity, by practically democratizing knowledge, and by organizing with technoscience for alternative worlds of ecological life.

But one could object here that craft as effective political organizing seems to be untenable when scale is at stake. How can alterontologies work with regard to the vast magnitude and impact of mainstream research in technoscience? How can alterontologies contribute to a decolonial politics of matter that develops alternative practices of justice within and against the frontier of matter? Following the earlier example, can grassroots ecological activism ever develop a viable alternative to big agribusiness and the technologies that it deploys? Infrastructural changes or large technoscientific projects require extensive mobilization in order to be open to democratic politics, and craft seems to be unable to achieve that; big science is too big to change under the pressure of alterontologies, as the argument goes. But is not the question about *scale-making*²⁶ the real issue here, rather than the entrancing size of technoscience? Craft is about rescaling the geographies of technoscience in ways that matter.

Consider the free software movement.²⁷ What mattered in the birth of the free software movement was not primarily the question of scale in terms of its size (that is, if free software could become so widespread as to potentially challenge the domination of proprietary software) but the attempt to change the conditions in which software was made. The primary goal of the movement was to allow for alternative forms of software creation to emerge: How is software owned (or not owned)? How can software become freely distributed? How can it efficiently engage a large number of actors in the global virtual space? How can peer-to-peer nets be set up? How can the gift economy of cyberspace be defended and maintained?—all questions that indicate that crafting new code and new forms of cooperation was not a strategy to attack proprietary software but to create livable digital worlds. The creation of alterontologies is not pri-

marily about size but about remaking the scale, not only crafting objects and processes but crafting scales too—that is, changing the values that a scale measures and not just its ratio, or else changing the scale itself. Scale-making in alterontologies is a *minoritarian* move: it focuses on the intensity of the actual moment of crafting rather than the extensity of the critique of mainstream technoscience. And as I discuss later in this book (in chapter 4 and in particular chapter 8), the relation between mainstream technoscience and alterontological practice is not clear-cut and oppositional, but characterized by multidirectional traffic, arduous exchanges, and endless collaborations and conflicts.

If a decolonial politics of matter is possible, then craft and artisanal skill lies in the ability to recognize the constraints of a situation, where to stop, and how to stop. Craft is about caring for the worlds we live in by acting in accordance with the intensities and the limits that matter imposes in each concrete situation. Maria Puig de la Bellacasa (2017) approaches care not as a moralistic stance but as a practical everyday engagement with the mundane worlds we inhabit that seeks to create more just and livable worlds. This is a generic notion of care: it can shape different organizational ontologies, but it is not universal. It needs to be materialized anew in each situation. In fact, the one who crafts alterontologies knows what the constraints are, where to stop, and how to leave himself or herself behind. Craft is not about *DIY*, but about *DIWY*: do it without yourself. Craft at its core is not about making things or producing relations but about leaving yourself aside for the sake of viably coexisting with other things and beings.